[INCH-POUND]
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SUPERSEDING
MIL-S-80086D
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COMMERCIAL ITEM DESCRIPTION

SHEARING MACHINES, METAL SQUARING, HYDRAULILC DRIVE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-S-80086D for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers hydraulically driven, metal squaring, shearing machines with equipment and accessories as described herein.

1.2 Intended use. Machines covered by this CID are Intended for use in metal shearing and squaring operations that are generally required in all phases of metal fabrication operations. Shearing machines having a gap frame can also be used for slitting, notching, and angular shearing operations.

2. CLASSIFICATION

2.1 Classification. Shearing machines shall be of the following types, sizes, and grades. The type, size, and grade to be furnished shall be as specified (see 7.2).

2.2 Type. Type I - Gapless/box frame
Type II - Gap frame

2.3 Size. Size is determined by material thickness and cutting length capacities as shown in Tables I and II.

2.4 Grade. Grade A - Machine deflection criteria as shown in Tables I and II.
Grade B - Machine deflection criteria as shown in Tables I and II.

Beneficial comments, recommendations, additions, deletions, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

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3 SALIENT CHARACTERISTICS

3.1 General requirements. The shearing machine shall be designed and manufactured in ;wcor-dance with the requirements of NFPA 79 and ANSI B11.4. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein The machine shall include all standard equipment, components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage. or distortion shall be accessible for adjustment, replacement, and repair. The machine shall be a self-contained power driven unit designed to accomplish shearing based on the fundamental principle of the downward movement of a ram mounted upper shear blade (knife) past a stationary lower shear blade. The machine shall have a means to permit leveling and mounting to the floor or foundation. The type I machine shall be provided with side housings of the gapless or straight design. Side housings for the type II machine shall have a gap or workpiece opening as specified (see 7.2) Unless otherwise specified (see 7.2), the paint color and application shall be in accordance with the manufacturer's standard practice.

3.2 Measurement systems. Either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable. the particular system required shall be as specified (see 7.2).

3.2.1 Measuring and indicating device. Either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable. the particular graduation required shall be as specified (see 7.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system When specified (see 7.2), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.3 Table. The table shall be of a one-piece construction with hand slots and T-slots machined into the table to accommodate front gauging. The T-slots shall meet the dimensional tolerances specified in section 4.5 of ASME B5.56M. The table shall securely support the lower shear blade and shall have a means for handling workplaces up to the rated capacity and rated strokes per minute of the machine without causing damage to the workpiece. Unless otherwise specified (7.2), the deflection of the table, during maximum shearing capacity, shall not be greater than the quantity specified in Tables I and II for the type, size, and grade of machine indicated. The table top shall have a surface finish of 125 micro inches or better and a flatness tolerance of +/- .002 in/ft or better when tested in accordance with ANSI/ASME B46.1. The distance from the floor to the top of the table shall not be less than 30 inches and not greater than 42 inches. The table on machine sizes 1/2 in and greater shall be provided with a self-cleaning ball transfer arrangement consisting of steel balls of not less than 3/4 in diameter.

3.4 Ram/crosshead and hold-down systems. The ram/crosshead and hold-down systems shall be powered by hydraulics and shall have the capability to shear the rated material capacity for the type, size, and grade of machine indicated in Tables I and II. The ram/crosshead hydraulic system shall evenly distribute the shearing force across the full length of the ram stroke. The hydraulic system shall conform to ANSI T2.24.1. Unless otherwise specified (see 7.2), the gib clearance

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TABLE I Machine sizes and performance requirements (Grade A), inches

Machine Size	Shearing Capacity		Shearing Accuracy (Drop) (Max, Shearing Capacity)			Machine Deflections (Max, Shearing Capacity)				Strokes
Material (Thickness x Width)	ASTM A36 80,000 psi Thickness	Stainless Steel (300 Series) 100,000 psi Thickness	Straightness	Parallelism	Squareness	Ram	Bed	Gib	Gib Clearance	Per Minute
1/4 x 72	1/4	3/16	.002	.002	.002	.001	.001	.001	.002	20
1/4 x 96	1/4	3/16	.002	.002	.002	.002	.002	.001	.002	20
1/4 x 120	1/4	3/16	.002	.002	.002	.002	.002	.001	.002	15
1/4 x 144	1/4	3/16	.002	.002	.002	.002	.002	.001	.002	13
1/4 x 168	1/4	3/16	.002	.002	.002	.002	.002	.001	.002	13
1/4 x 192	1/4	3/16	.002	.002	.002	.003	.003	.001	.002	13
1/4 x 240	1/4	3/16	.002	.002	.002	.004	.004	.001	.002	12
5/16 x 72	5/16	1/4	.002	.002	.002	.001	.001	.001	.002	20
5/16 x 96	5/16	1/4	.002	.002	.002	.002	.002	.001	.002	20
5/16 x 120	5/16	1/4	.002	.002	.002	.002	.002	.001	.002	15
5/16 x 144	5/16	1/4	.002	.002	.002	.002	.002	.001	.002	13
5/16 x 168	5/16	1/4	.002	.002	.002	.002	.002	.001	.002	13
5/16 x 192	5/16	1/4	.002	.002	.002	.003	.003	.001	.002	13
5/16 x 240	5/16	1/4	.002	.002	.002	.004	.004	.001	.002	12
3/8 x 72	3/8	5/16	.002	.002	.002	.002	.002	.001	.002	17
3/8 x 96	3/8	5/16	.002	.002	.002	.002	.002	.001	.002	13
3/8 x 120	3/8	5/16	.002	.002	.002	.002	.002	.001	.002	11
3/8 x 144	3/8	5/16	.002	.002	.002	.003	.003	.001	.002	10
3/8 x 168	3/8	5/16	.002	.002	.002	.003	.003	.001	.002	8
3/8 x 192	3/8	5/16	.002	.002	.002	.004	.004	.001	.002	8
3/8 x 240	3/8	5/16	.002	.002	.002	.005	.005	.001	.002	7
1/2 x 72	1/2	3/8	.002	.002	.002	.003	.003	.001	.002	13
1/2 x 96	1/2	3/8	.002	.002	.002	.003	.003	.001	.002	10
1/2 x 120	1/2	3/8	.002	.002	.002	.004	.004	.001	.002	9
1/2 x 144	1/2	3/8	.002	.002	.002	.005	.005	.001	.002	8
1/2 x 168	1/2	3/8	.002	.002	.002	.005	.005	.001	.002	7
1/2 x 192	1/2	3/8	.002	.002	.002	.006	.006	.001	.002	7
1/2 x 240	1/2	3/8	.002	.002	.002	.007	.007	.001	.002	6
5/8 x 72	5/8	1/2	.004	.004	.002	.003	.003	.002	.003	10
5/8 x 96	5/8	1/2	.004	.004	.002	.004	.004	.002	.003	9
5/8 x 120	5/8	1/2	.004	.004	.002	.006	.006	.002	.003	7
5/8 x 144	5/8	1/2	.004	.004	.002	.007	.007	.002	.003	6
5/8 x 168	5/8	1/2	.004	.004	.002	.008	.008	.002	.003	6
5/8 x 192	5/8	1/2	.004	.004	.002	.009	.009	.002	.003	5
5/8 x 240	5/8	1/2	.004	.004	.002	.010	.010	.002	.003	5

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TABLE I Machine sizes and performance requirements (Grade A), inches - Continued.

Machine Size Material (Thickness x Width)	Shearing Capacity		Shearing Accuracy (Drop) (Max, Shearing Capacity)			Machine Deflections (Max, Shearing Capacity)				Strokes Per Minute
	ASTM A36 80,000 psi Thickness	Stainless Steel (300 Series) 100,000 psi Thickness	Straightness	Parallelism	Squareness	Ram	Bed	Gib	Gib Clearance	
3/4 x 72	3/4	5/8	.008	.004	.003	.003	.005	.002	.003	10
3/4 x 96	3/4	5/8	.008	.004	.003	.005	.005	.002	.003	9
3/4 x 120	3/4	5/8	.008	.004	.003	.007	.007	.002	.003	7
3/4 x 144	3/4	5/8	.008	.004	.003	.008	.008	.002	.003	6
3/4 x 168	3/4	5/8	.008	.004	.003	.010	.010	.002	.003	6
3/4 x 192	3/4	5/8	.008	.004	.003	.011	.011	.002	.003	5
3/4 x 240	3/4	5/8	.008	.004	.003	.012	.012	.002	.003	5
1 x 72	1	3/4	.010	.004	.003	.005	.005	.002	.003	7
1 x 96	1	3/4	.010	.004	.003	.007	.007	.002	.003	6
1 x 120	1	3/4	.010	.004	.003	.009	.009	.002	.003	5
1 x 144	1	3/4	.010	.004	.003	.011	.011	.002	.003	4
1 x 168	1	3/4	.010	.004	.003	.013	.013	.002	.003	4
1 x 192	1	3/4	.010	.004	.003	.015	.015	.002	.003	3
1 x 240	1	3/4	.010	.004	.003	.017	.017	.002	.003	3
1-1/4 x 72	1-1/4	-	.010	.004	.003	.005	.005	.004	.003	4
1-1/4 x 96	1-1/4	-	.010	.004	.003	.008	.008	.004	.003	4
1-1/4 x 120	1-1/4	-	.010	.004	.003	.011	.011	.004	.003	3
1-1/4 x 144	1-1/4	-	.010	.004	.003	.013	.013	.004	.003	3
1-1/4 x 168	1-1/4	-	.010	.004	.003	.015	.015	.004	.003	2
1-1/4 x 192	1-1/4	-	.010	.004	.003	.017	.017	.004	.003	2
1-1/4 x 240	1-1/4	-	.010	.004	.003	.020	.020	.004	.003	2
1-1/2 x 72	1-1/2	-	.010	.004	.003	.005	.005	.005	.003	3
1-1/2 x 96	1-1/2	-	.010	.004	.003	.010	.010	.005	.003	3
1-1/2 x 120	1-1/2	-	.010	.004	.003	.013	.013	.005	.003	2
1-1/2 x 144	1-1/2	-	.010	.004	.003	.016	.016	.005	.003	2
1-1/2 x 168	1-1/2	-	.010	.004	.003	.019	.019	.005	.003	2
1-1/2 x 192	1-1/2	-	.010	.004	.003	.022	.022	.005	.003	2
1-1/2 x 240	1-1/2	-	.010	.004	.003	.025	.025	.005	.003	2

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TABLE II Machine sizes and performance requirements (Grade B), inches.

Machine Size (Material (Thickness x Width)	Shearing Capacity		Shearing Accuracy (Drop) (Max. Shearing Capacity)			Machine Deflections (Max. Shearing Capacity)				Strokes Per Minute
	ASTM A36 80,000 psi Thickness	Stainless Steel (300 Series) 100,000 psi Thickness	Straightness	Parallelism	Squareness	Ram	Bed	Gib	Gib Clearance	
1/4 x 72	1/4	3/16	.002	.002	.002	.002	.002	.002	.003	20
1/4 x 96	1/4	3/16	.002	.002	.002	.002	.002	.002	.003	20
1/4 x 120	1/4	3/16	.002	.002	.002	.002	.002	.002	.003	15
1/4 x 144	1/4	3/16	.002	.002	.002	.002	.002	.002	.003	13
1/4 x 168	1/4	3/16	.002	.002	.002	.002	.002	.002	.003	13
1/4 x 192	1/4	3/16	.002	.002	.002	.003	.003	.002	.003	13
1/4 x 240	1/4	3/16	.002	.002	.002	.004	.004	.002	.003	12
5/16 x 72	5/16	1/4	.002	.002	.002	.002	.002	.002	.003	20
5/16 x 96	5/16	1/4	.002	.002	.002	.002	.002	.002	.003	20
5/16 x 120	5/16	1/4	.002	.002	.002	.002	.002	.002	.003	15
5/16 x 144	5/16	1/4	.002	.002	.002	.002	.002	.002	.003	13
5/16 x 168	5/16	1/4	.002	.002	.002	.002	.002	.002	.003	13
5/16 x 192	5/16	1/4	.002	.002	.002	.003	.003	.002	.003	13
5/16 x 240	5/16	1/4	.002	.002	.002	.004	.004	.002	.003	12
3/8 x 72	3/8	5/16	.002	.002	.002	.003	.002	.002	.003	17
3/8 x 96	3/8	5/16	.002	.002	.002	.004	.003	.002	.003	13
3/8 x 120	3/8	5/16	.002	.002	.002	.004	.003	.002	.003	11
3/8 x 144	3/8	5/16	.002	.002	.002	.005	.004	.002	.003	10
3/8 x 168	3/8	5/16	.002	.002	.002	.005	.004	.002	.003	8
3/8 x 192	3/8	5/16	.002	.002	.002	.006	.005	.002	.003	8
3/8 x 240	3/8	5/16	.002	.002	.002	.006	.005	.002	.003	7
1/2 x 72	1/2	3/8	.002	.002	.002	.005	.005	.002	.003	13
1/2 x 96	1/2	3/8	.002	.002	.002	.005	.005	.002	.003	10
1/2 x 120	1/2	3/8	.002	.002	.002	.006	.006	.002	.003	9
1/2 x 144	1/2	3/8	.002	.002	.002	.007	.007	.002	.003	8
1/2 x 168	1/2	3/8	.002	.002	.002	.007	.007	.002	.003	7
1/2 x 192	1/2	3/8	.002	.002	.002	.008	.008	.002	.003	7
1/2 x 240	1/2	3/8	.002	.002	.002	.009	.009	.002	.003	6
5/8 x 72	5/8	1/2	.004	.004	.002	.005	.005	.003	.003	10
5/8 x 96	5/8	1/2	.004	.004	.002	.006	.006	.003	.003	9
5/8 x 120	5/8	1/2	.004	.004	.002	.008	.008	.003	.003	7
5/8 x 144	5/8	1/2	.004	.004	.002	.009	.009	.003	.003	6
5/8 x 168	5/8	1/2	.004	.004	.002	.010	.010	.003	.003	6
5/8 x 192	5/8	1/2	.004	.004	.002	.011	.011	.003	.003	5
5/8 x 240	5/8	1/2	.004	.004	.002	.012	.012	.003	.003	5

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TABLE II Machine sizes and performance requirements (Grade B), inches - Continued

Machine Size	Shearing Capacity		Shearing Accuracy (Drop) (Max, Shearing Capacity)			Machine Deflections (Max, Shearing Capacity)				Strokes Per Minute
Material (Thickness x Width)	ASTM A36 80,000 psi Thickness	Stainless Steel (300 Series) 100,000 psi Thickness	Straightness	Parallelism	Squareness	Ram	Bed	Gib	Gib Clearance	
3/4 x 72	3/4	5/8	.008	.004	.003	.005	.005	.003	.003	10
3/4 x 96	3/4	5/8	.008	.004	.003	.007	.007	.003	.003	9
3/4 x 120	3/4	5/8	.008	.004	.003	.009	.009	.003	.003	7
3/4 x 144	3/4	5/8	.008	.004	.003	.010	.010	.003	.003	6
3/4 x 168	3/4	5/8	.008	.004	.003	.012	.012	.003	.003	6
3/4 x 192	3/4	5/8	.008	.004	.003	.013	.013	.003	.003	5
3/4 x 240	3/4	5/8	.008	.004	.003	.014	.014	.003	.003	5
1 x 72	1	3/4	.010	.004	.003	.007	.007	.003	.003	7
1 x 96	1	3/4	.010	.004	.003	.010	.010	.003	.003	6
1 x 120	1	3/4	.010	.004	.003	.012	.012	.003	.003	5
1 x 144	1	3/4	.010	.004	.003	.015	.015	.003	.003	4
1 x 168	1	3/4	.010	.004	.003	.017	.017	.003	.003	4
1 x 192	1	3/4	.010	.004	.003	.018	.018	.003	.003	3
1 x 240	1	3/4	.010	.004	.003	.019	.019	.003	.003	3
1-1/4 x 72	1-1/4	-	.010	.004	.003	.007	.007	.004	.003	4
1-1/4 x 96	1-1/4	-	.010	.004	.003	.010	.010	.004	.003	4
1-1/4 x 120	1-1/4	-	.010	.004	.003	.018	.018	.004	.003	3
1 1/4 x 144	1 1/4	-	.010	.004	.003	.020	.020	.004	.003	3
1-1/4 x 168	1-1/4	-	.010	.004	.003	.023	.023	.004	.003	2
1-1/4 x 192	1-1/4	-	.010	.004	.003	.024	.024	.004	.003	2
1 1/4 x 240	1-1/4	-	.010	.004	.003	.025	.025	.004	.003	2
1-1 2 x 72	1-1/2	-	.010	.004	.003	.007	.007	.005	.003	3
1-1 2 x 96	1-1/2	-	.010	.004	.003	.011	.011	.005	.003	3
1-1/2 x 120	1-1/2	-	.010	.004	.003	.019	.019	.005	.003	2
1-1/2 x 144	1-1/2	-	.010	.004	.003	.022	.022	.005	.003	2
1-1/2 x 168	1-1/2	-	.010	.004	.003	.025	.025	.005	.003	2
1-1/2 x 192	1-1/2	-	.010	.004	.003	.028	.028	.005	.003	2
1-1/2 x 240	1-1/2	-	.010	.004	.003	.030	.030	.005	.003	2

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3.5 Lubrication. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible. Shearing machines requiring lubrication if the ram/crosshead drive shall be provided with either a manual-one shot or automatic lubrication system (see 7.2) and shall have a replaceable in-line filter capable of removing particles 10 microns in size. A means for determining lubricant level in the reservoir(s) shall be by sight gauge. warning light, or a similar device.

3.6 Shear baldes/knifes. Unless otherwise specified (see 7.2), the manufacturer's standard commercial blades shall be provided. The shear blades shall be reversible and shall have four cutting edges. Unless otherwise specified (see 7.2). the blades shall comply with the dimensional tolerances of ASME B94.53 and section 5 and Table 3 of ASME B5.56M. Blade parallelism shall be in accordance with section 7.2.1 and Table 4 of ASME B5.56M.

3.7 Gauging. The machine shall be provided with an accurate means for aligning and positioning the workpiece prior to shearing The machine shall be provided with front, back, and side (squaring-arm) gauging.

3.7.1 Back gauge. A back gauge shall be provided on the rear of the machine and shall provide an adjustable reference edge between the back edge of the workpiece and the lower shear blade. The back gauge shall be designed to prevent the workpiece from being jammed between the backstop and the lower blade during shearing. Parallelism of the back gauge to the lower blade shall be in accordance with section 7.2.2 of ASME B5.56M. The back gauge shall be power operated, programmable, or manual as specified (see 7.2). Powered and programmable back gauges shall have a traverse speed of not less than 60 in/min. The back gauge shall permit shearing of workpieces whose dimensions exceed the back gauge capacity and shall have a means for dampening impact loads. Unless otherwise specified (see 7.2), the back gauge shall have a workpiece capacity of not less than 36 in.

3.7.2 Front gauge. The front gauge shall provide an adjustable reference edge between the front edge of the workpiece and the lower shear blade. The front gauge shall consist of not less than two support arms, a recessed precision stainless steel scale graduated in not more than 1/32 in. increments, and two disappearing type, adjustable work stops. The front gauge shall be power operated or manual as specified (see 7.2). Unless otherwise specified (see 7.2), the front gauge shall have a workpiece capacity of 48 in. When specified (see 7.2), front gauging shall be provided on machines rated over 1/2 in. capacity.

3.7.3 Side squaring gauge The machine shall be provided with a side squaring gauge to position the edge of the workpiece at 90 degrees to the lower shear blade. The squaring gauge shall be capable of interfacing to either the right or left side of the machine and shall have a graduated scale

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calibrated in not more than 1/32 in. increments. The side squaring gauge shrill be power operated or manual as specified (see 7.2) Unless otherwise specified (see 7.2), the side gauges shall have a workpiece capacity of 48 in. When specified (see 7.2), an extension side gauge, with length as specified shall be provided with the machine.

3.7.4 Optional gauging. When specified (see 7.2), the following optional gauges shall be provided:

a. Shadow light.
b. Angle gauge.

3.8 Controls. All controls and electronic displays shall be located convenient to the operator's work station(s).

3.8.1 Computer numerical control (CNC). When specified (see 7.2), the machine shall be provided with a solid-state CNC system for positioning of the back gauge. When specified (see 7.2), the CNC system shall provide programmable control of the front gauge and side gauges. The CNC system shall be capable of sequential programming having not less than 15 different gauge settings (front or back) and not less than 99 cuts per setting. The CNC system shall have the capability to operate in either English or metric units and shall be provided with not less than the following features.

a. A manual and automatic mock of operation.
b. Automatic return to any previous step in a sequence.
c. Display and track the step. gauge position. and the current number of cuts in the sequence.
d. Self-diagnostic.
e. Battery back-up.

3.8.2 Ram/crosshead drive. The machine shall have controls for operating the ram in both single stroke and continuous stroke operations and for advancing the ram to check blade clearance adjustment. The drive system shall be operated by a remote control foot switch.

3.8.3 Blade clearance adjustment. The ram shall have provisions for either manual, powered, or programmable control of' blade clearance adjustment (see 7.2). The blade clearance adjustment positioning accuracy shall be +/- .005 in. of target setting.

3.8.4 Ram stroke adjustment. The ram shall have provisions for either manual, powered, or programmable control of the ram stroke adjustment (see 7.2). Unless otherwise specified (see 7.2), the ram stroke shall be adjustable from 24 in up to full stroke capacity

3.8.5 Blade straightening adjustment. The ram shall have provisions for manually straightening the blade to comply with the parallelism tolerances as specified m section 7.2.1 of ASME B5.56M.

3.8.6 Rake angle adjustment. Machines with an adjustable rake angle shall have provisions for either manual, powered, or programmable control of the rake angle (see 7.2).

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3.8.7 Gauge adjustments Unless otherwise specified (see 7.2), powered or programmable gauges shall have a positioning accuracy to within +/- .003 in.. of target setting.

3.9 Electrical requirements. The machine's electrical system shall conform to NFPA 79 and Article 430, Part C of NFPA 70. The machine shall operate on dual 230/460 volt, 3-phase, 60 Hertz electrical power. Unless otherwise specified (see 7.2), the machine shall be initially wired for 240 volt operation. All motors shall be energy efficient and shall conform to the requirements of NEMA MG-1 .

3.10 Performance standards. The machine shall be capable of shearing ASTM A36 steel of not less than 80.000 psi (550 MPa) tensile strength in workpiece sizes ranging up to the rated capacity for the type and size machine specified in Tables I and II. The shearing accuracy's (straightness, parallelism, squareness) of the workpiece drops and the deflection of the ram, bed, and gibs shall not be greater than the quantities specified in Tables I and II for the type, size, and grade of machine indicated.

3.11 Safety and health requirements. The machine and all associated equipment and accessories shall be in compliance with ANSI B11.4, ANSI B11.19, ANSI B15.1, OSHA 1910.95, OSHA 1910.212, and OSHA 1910.219. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.12 Technical Information. All technical information/data shall be furnished as specified herein and with the requirements of this contract. All technical data shall be written in the English language.

3.12.1 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2), the nameplate shall contain the following information.

a. Nomenclature.
b. Manufacturer's name.
c. Machine model designation.
d. Power input (volts, total amps. phase, frequency).
c. Short circuit/over-current rating.
f. Manufacturer's serial no.
g. Shearing capacity at rated rake angle.
h. Contract number or order number.
i National stock number or plant equipment code.
j. Date of manufacture.

3.12.2 Lubrication plate or chart. When specified (see 7.2), a lubrication plate or chart shall be attached to the machine. The chart shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. The information provided on the plate or chart shall include the follow information.

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

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3.13 Standard equipment. The manufacturer shall furnish all equipment considered standard for the machine.

3.14 Optional equipment. Optional equipment shall be furnished as specified (see 7.2).

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4.2 Environmental protection. The machine shall meet all applicable Environmental Protection Agency (EPA) and occupational Safety and Health Administration restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health and shall be met during the manufacturing, service, transportation, storage, and operation of the machine.

5. QUALITY ASSURANCE PROVISIONS

5.1 Porduct conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market The government reserves the right to require proof of such conformance.

5.2 Examination. The machine shall be examined to determine compliance with all requirements specified in this CID.

5.3. Testing. Unless otherwise specified (see 7.2), the machine shall be tested in accordance with the following procedures to assure compliance with the safety and performance requirements specified herein, Preliminary and final acceptance testing shall be conducted at the origin (manufacturer's site).

5.3.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 5 degrees to 40 degrees Celsius at 20 to 95 percent humidity.

5.3.2 Operational test. The machine shall be operated at no load for not less than one hour. All controls, motors, adjusting mechanisms, and gauges shall be actuated and observed to determine compliance with the requirements specified herei.

5.3.3 Performance tests. Unless otherwise specified (see 7.2), the workpiece (material) shall be provided by the maunfacturer. The rake angle and blade gap shall not be greater than the quantity specified in Table III for the machine capacity indicated. The machine shall shear six workpieces at maximum rated capacity and six workpieces at 20 percent of the rated capacity. The machine shall be tested in accordance with section 7 of ASME B5.56M. Unless otherwise specified (7.2), the instruments and methods used in performing the tests for measuring gib clearance and deflection shall be at the option of the contractor/manufacturer, provided that they accurately determine the clearance and deflections of the machine as approved by the contracting officer. The performance accuracy's of the machine shall be in accordance with section 3.10.

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TABLE III. Machine settings, in.

Machine Capacity	Rake Angle	Blade Gap
0.250 in.	0 375 in	0.015
0 375 in.	0.437 in	0.022
0.500 in.	0.500 in.	0.030
0.625 in.	0.562 in.	0.037
0.750 in	0.625 in.	0 045
1.000 in.	0.750 in.	0.060
1.250 in.	0.812 in.	0.075
1 500 in.	0.875 in.	0.090

6. PACKAGING

6.1 Packaging requirements. Unless otherwise specified (see 7.2), the machine shall be packaged in accordance with the manufacturer's standard practices.

7. NOTES

71 Referenced documents.

7.1.1 Government documents. Copies of Military, Federal, and other government documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC.

7.1.2 Industry/commercial documents. Copies of industry/commercial documents may be obtained from the following addresses:

a. American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.
b. American Society of Mechanical Engineers (ASME), 345 E. 47th Street, New York, NY 10017
c National Fire Protection Association (NFPA), Batterymarch Park, Quincy, MA 02269
d National Electrical Manufacturers Association (NEMA), 2101 L Street Washington DC 20037.
e Occupational Safety and Health Association (OSHA) (See section 7.2.1)
f Joint Industrial Council (JIC), 7901 Westpark Drive, McLean, VA 22101

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7.2 Ordering data. Acquisition documents must specify the following:

a Title, number, and date of this document.
b Type, size, and class of machine required (see 2.0).
c Gap size (type II machine), specify (see 3.1).
d. Paint color, if different (see 3.1).
e Measurement system specify if required (see 3.2).
f Measurement system for indicating devices, specify if required (see 3.2.1).
g. Dual measurement system for indicating devices, if required (see 3.2.1).
h. Table deflection, if different (see 3.3).
i. Gib clearance, if different (see 3.4).
j. Ram and gib deflection, if different (see 3.4).
k Independent control of hold-down system, if required (see 3.4).
l Lubrication system, manual-one shot/automatic (see 3.5).
m Shear blades, indifferent (see 3.6).
n Shear blade dimensional tolerances, if different (see 3.6).
o Back gauge, specify manual, powered, or programmable (see 3.7.1).
p Back gauge capacity (length), if different (see 3.7.1).
q Front gauge, specify manual or powered (see 3.7.2).
r Front gauge capacity (length), if different (see 3.7.2).
s. Front gauge for machine rated over 1/2 in. shearing capacity, if required (see 3.7.2).
t. Side squaring gauge, specify manual or powered (see 3.7.3).
u. Side squaring gauge capacity (length), if different (see 3 .7.3).
v Side squaring gauge extension, if required (see 3.7.3).
w Optional gauges (i.e shadow light or angle gauge), specify if required (see 3.7.4)
x Computer numerical control (CNC), if required (see 3.8.1).
y CNC options (i.e Front and side sq. gauges, rake angle, etc.), if required (see 3.8.1).
z Blade clearance adjustment, specify manual or powered (see 3.8.3).
aa. Ram stroke adjustment, specify manual or powered (see 3.8.4).
bb Rake angle adjustment (if adjustable), specify manual or powered (see 3.8.6).
cc. Electrical power requirements, if different (see 3.9).
dd. Nameplate information, if different (see 3.12.1).
ee. Lubricant chart/plate, if required (see 3.12.2).
ff Optional equipment, as specified (see 3.14).
gg Testing, if different (see 5.3).
hh Workpiece/material used for testing, if different (see 5.3.3).
ii Testing instruments and methods, if different (see 5.3.3).
jj Packaging requirements, if different (see 6.1).

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7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

AA55805-XXXX

Grade	A - Grade A B - Grade B	
Size	1 - 72 in. 2 - 96 in. 3 - 120 in. 4 - 144 in.	5 - 168 in. 6 - 192 in. 7 - 240 in.
Capacity:	A - 1/4 in B - 5/16 in C - 3/8 in D - 1/2 in E - 5/8 in.	F - 3/4 in G - 1 in. H - 1-1/4 in I - 1-1/2 in.
Type	1 - Type I (Gapless/Box frame) 2 - Type II (Gap frame)	

MILITARY INTERESTS

Custodians
Army - AL
Air Force - 99

Reviewers
Air Force - 84
Navy - SH

CIVIL AGENCY COORDINATING ACTIVITY
GSA-FSS

PREPARING ACTIVITY:
DLA - GS

(Project 3445-0132)

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